Reg. No. | | | | | | | | | | | [13AE53]

FIFTH SEMESTER B.Tech/B.E EXAMINATION, DECEMEBER–2015

AIRCRAFT STUCTURES - II

Time: 3 Hours **Maximum Marks:70**

Instructions: Answer any five full questions.

1.	**a.**	State the assumptions made in deriving equilibrium equation for an Euler's column.	**02**
	b.	Derive Euler's load (P_{cs}) for the first two buckling loads and sketch the two corresponding mode shapes.	**04**
	c.	Prove that for a fixed-fixed (clamped-clamped) column the critical buckling load is four times the Euler's buckling load.	**08**
2.	**a.**	Define total potential of a structural system of a column with an load	**02**
	b.	Spet an expression for the total potential (Π_p) of a system consisting of a column with an axial load.	**04**
	c.	Assuming a suitable displacement function for a fixed pinned column obtain critical buckling load. (Fourth degree polynomial is preferable)	**08**
3.	**a.**	What are beam columns? Give two examples with sketches.	**04**
	b.	Derive an expression for maximum bending moment of a centrally loaded beam column.	**10**
4.	**a.**	State the assumptions made in deriving plate buckling equation.	**04**
	b.	Derive the governing differential equation of plate buckling under in-plane loads.	**10**
5.	**a.**	Explain different ways of stiffening of prates.	**04**
	b.	Deduce critical buckling load (stress) for a simply supported stiffened prate. (uni-axially loaded)	**10**
6.	**a.**	Explain the buckling behavior of monocoque and semi-monocoque shells used is aircraft structures.	**04**
	b.	Get an expression for critical load for a circular prate under inplane loads.	**10**
7.	**a.**	Explain post buckling behavior of full diagonal tension field web beam.	**04**
	b.	Obtain expressions for the wing spar beam under a tip load – diagonal tension load, flange load stiffener load.	**10**
8.		Figure below illustrates a circular fuselage with stiffeners marked as small circles. The following data are given:	**14**

- Area of each stringer = $96mm^2$
- Skin thickness =1mm
- All materials of Al. alloy with
- E = 7000 MP_a ; $\nu = 0.3$
- Fuselage frame spacing = 400mm
- Fuselage loads :

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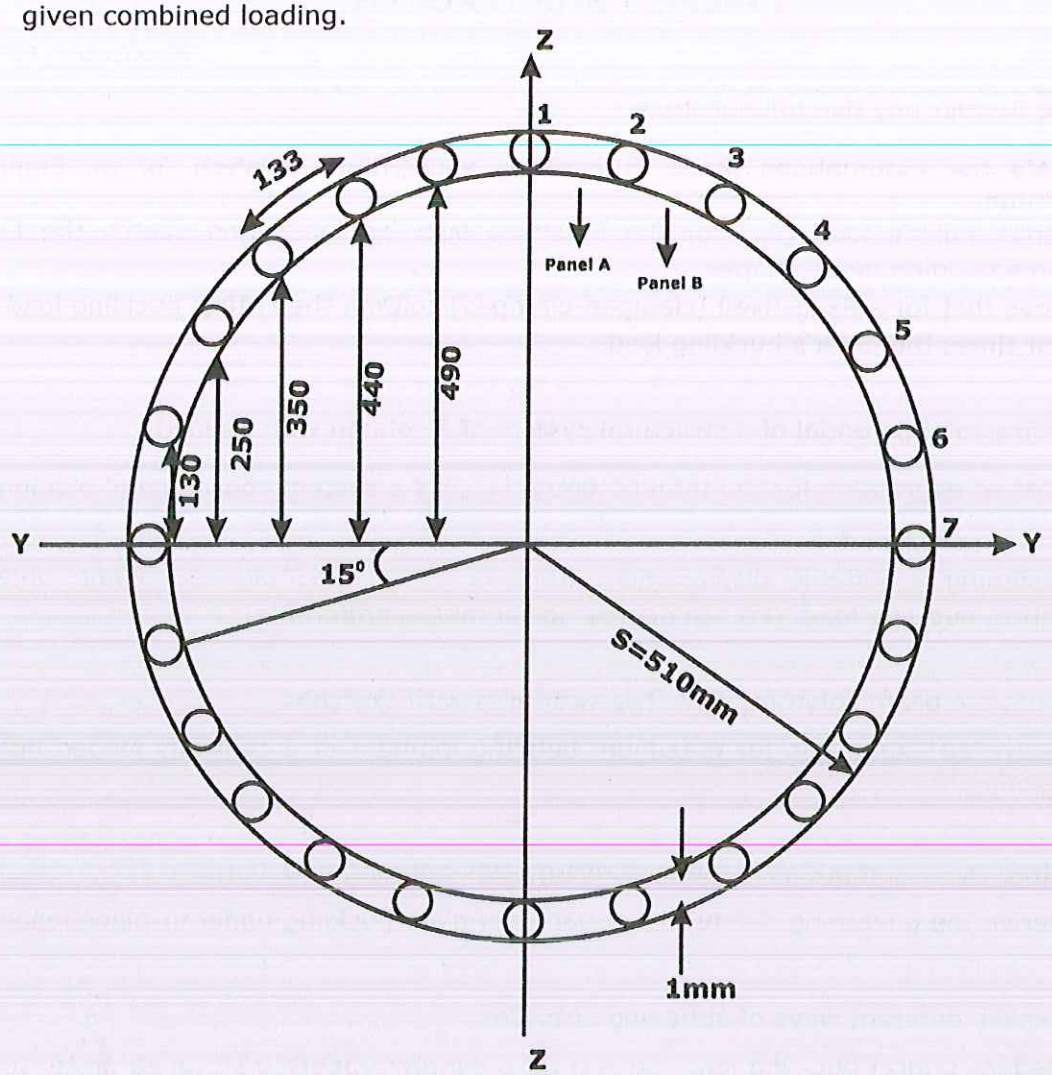
M_y = 67790 N.m (Bending Moment)
T = 23726 N.m (Tensional moment)

- Bending moment causing compression on top half
- Tensional moment acting center clockwise.

Determine whether the skin panels A and B of the fuselage will buckle under the given combined loading.